Jaypee Institute of Information Technology, Noida
T1 Examination, EVEN 2026
B.Tech Semester VI

Course Title: Computer Networks & Internet of Things **Max. Hours: 1 Hr**
Course Code: 18B11CS311 **Max. Marks: 20**

COURSE OUTCOMES

CO1	Defining the basics of networking, components and underlying technologies
CO2	Illustrate the various key protocols in OSI model and TCP/IP protocol suite and explain various application protocols.
CO3	Examine various transport protocols and its performance enhancing mechanisms.
CO4	Determine the shortest path for the network using various routing protocols and evaluate it.
CO5	Choose IP & MAC addressing mechanisms and data link layer protocols to solve communication, error detection and correction problems.
CO6	Identification and description of various components, architectures and protocols of Internet of Things (IoT) and their real-life problems.

Note: Attempt all questions.

Q.1. A client accesses a web page with a 20 KB HTML file and four embedded objects (Image 1: 50 KB, Image 2: 100 KB, CSS: 10 KB, JS: 30 KB). Embedded objects are requested only after the HTML file is fully received. RTT = 2 s, server TCP window = 50 KB, and link bandwidth is high (window-limited). Calculate total retrieval time for different cases.

a) HTTP/1.0 without parallel connections.
b) HTTP/1.0 with up to two parallel connections,
c) HTTP/1.1 without pipelining
d) HTTP/1.1 with pipelining.

[CO2(Understanding), 4 Marks]

Q.2. A client machine at student.cse.jiit.ac.in wants to access the remote host **www.cs.standford.edu**, which has the canonical name **portal.course.cs.stanford.edu** and IP address **185.45.34.23**. DNS caching is disabled. The DNS resolution involves various RTTs: 10 ms between the client and local DNS server, 20 ms from local to root DNS, 25 ms from root to TLD DNS, and 30 ms from TLD to authoritative DNS. Direct RTTs are 23 ms from local to TLD and 15 ms from local to authoritative. Each DNS reply incurs a transmission delay of 5 ms.

a) Describe the sequence of iterative DNS query–response steps for resolving www.cs.stanford.edu, indicating the type and number of Resource Records (NS, A, CNAME) returned at each stage.
b) Calculate the total time required for iterative DNS resolution using the given RTTs, and list the Resource Records returned by the authoritative DNS server, clearly showing all field values.

[CO2(Understanding), 4 Marks]

Q.3. Answer the following:

a) A computer in the IT department sends data to a Finance department computer through 2 switches and 2 routers using TCP/IP. Assuming one-way transmission and no fragmentation, determine how many times each TCP/IP layer is used during the transfer.

b) Given the TCP header dump **30 39 00 50 00 00 03 E8 00 00 00 00 50 20 20 00 00 00 01 2C**, determine sequence number range for urgent data (if any) and the header length.

[CO (Remembering), CO3 (Analysing), 4 Marks]

Q.4. A multinational IT company is transferring a 10 MB (Mega Bytes) backup file (divided into 10 packets of 1 MB) from its main server (S) to a cloud storage center (D) over a packet-switched WAN. The network uses store-and-forward switching and dynamic routing. All links in the network have the same distance of 100 km, and the propagation speed on each link is 2×10^8 m/s. Each router adds 1 ms processing delay per packet. Due to traffic, packets take different paths through interconnected routers. (in Fig 1)

- **Packets 1–5 (Primary Route):** S → R1 → R2 → R3 → R6 → R8 → D
- **Packets 6–10 (Secondary Route):** S → R1 → R4 → R5 → R7 → R2 → R3 → R6 → R8 → D

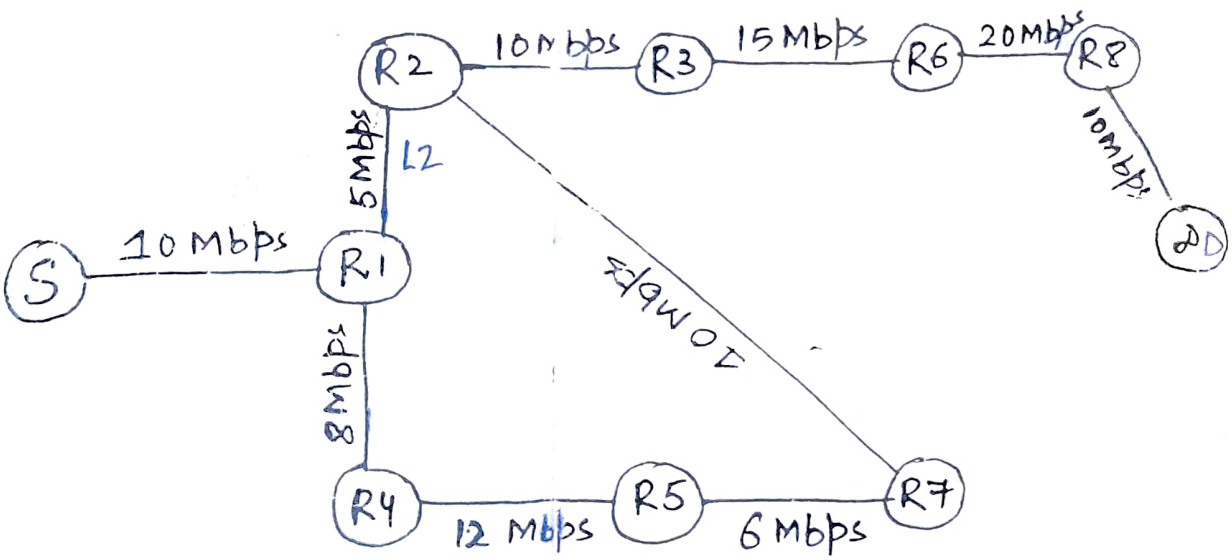

Fig 1. Network

a) How many bits can accommodate on the Link L2 at any given time.

b) Is there any time lag (in seconds) between packet 3 and 6 received at Route R6. If yes, how much is the time Lag?

c) Find the total end-to-end delay (in seconds) of all the packets to reach destination D.

d) Calculate the achievable throughput of the connection (in Mbps).

e) Assume that the bottleneck link on the primary route employs TDM with 5 equal time slots allocating one slot per packet flow. Determine the effective bandwidth available per flow(in Mbps) and the new transmission delay per packet on bottleneck link (in seconds).

[CO1(Remembering), 1+1+3+1+2=8 Marks]